

A Fredholm-index obstruction to essential unitary equivalence

A negative answer to Question 6.4 of arXiv:2407.18455

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Candidate full counterexample. The two extensions in Question 6.4 need not be essentially unitarily equivalent. In the specialization $p(z, w) = z - w$, $B_1 = B_2 = \text{id}$, the second extension is the Bergman Toeplitz extension and its coordinate lift has Fredholm index -1 . For the first extension, the corresponding coordinate lift is invertible and has index 0 . Essential unitary equivalence would preserve this index.

Source question and transcription

The source chooses a distinguished homogeneous polynomial $p = \prod_i (z - \alpha_i w)^{n_i}$, finite Blaschke products B_1, B_2 , and an infinite Blaschke product B whose zeros accumulate at every point of $\mathbb{T}$. It sets

$$B_3 = B \circ B_1, \quad B_4 = \prod_i B \circ (\alpha_i B_2), \quad \mathcal{M} = [B_3(z)B_4(w), p_B].$$

The paper proves that $\mathcal{M}^\perp$ and $[p_B]^\perp$ are essentially normal and have the same essential spectrum, hence produce two extensions of the same commutative C^* -algebra by the compact operators. Question 6.4 asks whether these extensions are essentially unitarily equivalent.

Definitions

Two quotient-module extensions over $C(X)$ are *essentially unitarily equivalent* if a unitary between their Hilbert spaces conjugates the lifts of each coordinate function modulo compact operators. In particular, whenever a coordinate lift minus λ is Fredholm, its index is invariant under such an equivalence.

The Bergman space $A^2(\mathbb{D})$ has orthonormal basis $\{\sqrt{n+1}\zeta^n : n \geq 0\}$. Multiplication by ζ is the weighted shift with weights $\sqrt{(n+1)/(n+2)}$; it is injective with one-dimensional cokernel and therefore has index -1 .

Idea / proof intuition

Collapse the bidisc quotient by $z - w$ onto the diagonal. The second quotient becomes the Bergman module. Adding $B(z)B(w)$ becomes quotienting the Bergman module by the invariant subspace generated by B^2 . If $B(0) \neq 0$, a one-line Bezout identity $a(z)z + B(0)^{-2}B(z)^2 = 1$ makes the new coordinate compression invertible. The resulting index jump obstructs essential unitary equivalence.

Original source image

then by [19, Lemma 2.2] and [21, Theorem 0.1], it is not difficult to verify

$$\sigma_e([p_B]^\perp) \subseteq \overline{Z(B_3(z)B_4(w)) \cap Z(p_B) \cap \partial\mathbb{D}^2} \subseteq \sigma_e(\mathcal{M}^\perp).$$

□

By Theorem 4.4 both $\mathcal{M}^\perp = [B_3(z)B_4(w), p_B]^\perp$ and $[p_B]^\perp$ are essentially normal, hence there are two C^* -extensions

$$0 \rightarrow \mathcal{K} \rightarrow C^*(M^\perp) \rightarrow C(\sigma_e([p_B]^\perp)) \rightarrow 0, \quad (6.1)$$

and

$$0 \rightarrow \mathcal{K} \rightarrow C^*([p_B]^\perp) \rightarrow C(\sigma_e([p_B]^\perp)) \rightarrow 0. \quad (6.2)$$

From Theorem 3.6 the short exact sequence (6.2) gives a nontrivial element in the K-homology group $K(\sigma_e([p_B]^\perp))$. We end this section by the following question.

Question 6.4. *Are these two C^* -extensions essentially unitarily equivalent?*

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References

Source: Wang-Zhao-Zhu, arXiv:2407.18455, PDF p. 16, equations (6.1)–(6.2) and Question 6.4. This is a direct crop from source_paper.pdf.

The specialized pair of quotient modules

For $m \geq 2$ and $0 \leq k < m$, put

$$\beta_{m,k} = r_m e^{2\pi i k/m}, \quad r_m = 1 - \frac{2^{-m}}{m^2}.$$

Then

$$\sum_{m=2}^{\infty} \sum_{k=0}^{m-1} (1 - |\beta_{m,k}|) = \sum_{m=2}^{\infty} \frac{2^{-m}}{m} < \infty.$$

Hence these points are the zeros of an infinite Blaschke product B . Their boundary cluster set is all of $\mathbb{T}$, while $B(0) \neq 0$: the angular grids become dense, and convergence of the displayed sum makes the product of the zero moduli at the origin strictly positive.

Take

$$p(z, w) = z - w, \quad B_1(z) = z, \quad B_2(w) = w.$$

This is an admissible instance of the source construction, with

$$p_B = z - w, \quad B_3 = B_4 = B,$$

and quotient modules

$$\mathcal{N}_0 = [z - w]^\perp, \quad qquad \mathcal{N}_1 = [z - w, B(z)B(w)]^\perp.$$

By the source result, both are essentially normal and $\sigma_e(\mathcal{N}_0) = \sigma_e(\mathcal{N}_1) = \{(\zeta, \zeta) : \zeta \in \mathbb{T}\}$, which we identify with $\mathbb{T}$.

Lemma 1 (Diagonal Bergman model). *There is a unitary $U : \mathcal{N}_0 \rightarrow A^2(\mathbb{D})$ under which both compressed coordinate multipliers become M_ζ , and the compression of $B(z)B(w)$ becomes M_{B^2} . Moreover,*

$$U\mathcal{N}_1 = K := A^2(\mathbb{D}) \ominus \overline{B^2 A^2(\mathbb{D})}.$$

Proof. The vectors

$$e_n = \frac{1}{\sqrt{n+1}} \sum_{j=0}^n z^j w^{n-j}, \quad n \geq 0,$$

form an orthonormal basis of $[z - w]^\perp$. Diagonal restriction sends e_n to $\sqrt{n+1} \zeta^n$, so it defines the asserted unitary. Projection of either ze_n or we_n onto $\mathcal{N}_0$ is $\sqrt{(n+1)/(n+2)} e_{n+1}$. The bounded analytic functional calculus then sends $B(z)B(w)$ to $B(M_\zeta)^2 = M_{B^2}$. Finally, quotienting by the closed submodule generated by this multiplier gives the stated orthogonal complement. This complement is infinite-dimensional, since the Bergman reproducing kernels at the infinitely many zeros of B belong to it and are linearly independent. $\square$

Index obstruction

Let $T = P_K M_\zeta|_K$ be the coordinate compression on K .

Theorem 1. *For the above admissible source data, the extensions associated with $\mathcal{N}_1$ and $\mathcal{N}_0$ are not essentially unitarily equivalent.*

Proof. Set $c = B(0)^{-2}$ and

$$a(z) = \frac{1 - cB(z)^2}{z}.$$

The apparent singularity at zero is removable. The resulting function is in H^∞ : it is bounded on $|z| \leq 1/2$ by compactness, and on $|z| \geq 1/2$ by the displayed quotient. Thus

$$a(z)z + cB(z)^2 = 1. \tag{1}$$

Because $\overline{B^2 A^2(\mathbb{D})}$ is invariant under every bounded analytic multiplier, compression to K defines a unital homomorphism $H^\infty \rightarrow B(K)$. The compressed multiplier by B^2 is zero. Applying this homomorphism to (1) gives

$$a(T)T = Ta(T) = I_K.$$

Hence T is invertible and $\text{ind } T = 0$.

On $\mathcal{N}_0$, the corresponding coordinate lift is unitarily equivalent to the Bergman shift M_ζ , which has $\text{ind } M_\zeta = -1$. If the two extensions were essentially unitarily equivalent, their coordinate lifts would be unitarily conjugate modulo compact operators. Fredholm index is unchanged by unitary conjugacy and compact perturbation, forcing $0 = -1$, a contradiction. $\square$

Stronger observation. The explicit radial-grid choice of B is only used to meet the source's dense boundary-zero hypothesis and ensure $B(0) \neq 0$. More generally, for any admissible B and any $\lambda \in \mathbb{D} \setminus Z(B)$, the same argument with $c = B(\lambda)^{-2}$ and $a(z) = (1 - cB(z)^2)/(z - \lambda)$ shows that $T - \lambda$ is invertible, whereas $M_\zeta - \lambda$ has index -1 .

Verification, novelty, and scope

- The Blaschke condition and full-circle boundary cluster set are checked directly from the radial grid.
- The diagonal basis calculation verifies the Bergman weights and the index -1 without invoking an abstract model theorem.
- The passage from $B(z)B(w)$ to B^2 uses the quotient-module homomorphism property, so no unproved equality of raw multiplier ranges is needed; closures are retained throughout.
- The Bezout coefficient a is explicitly bounded, avoiding any appeal to the corona theorem.
- Both left and right inverse identities hold because the compressed H^∞ calculus is commutative.

Novelty audit. Cheap run indexes were searched for arXiv:2407.18455, its exact final question, the two displayed extension formulas, and Bergman/model-space index obstructions. Bounded primary-source web/arXiv searches through 17 August 2026 found the source paper and later talks describing its main results, but no later paper or preprint resolving Question 6.4. Novelty confidence is moderate rather than definitive.

Limitations. This gives a complete negative answer to the universal reading of Question 6.4 by exhibiting an admissible instance in which equivalence fails. It does not classify when other choices of p, B_1, B_2, B yield equivalent extensions, and it does not resolve Yang’s separate Hilbert–Schmidt conjecture or the source’s conjecture about all infinitely generated submodules.

Human review. Pending. Priority checks are the identification of the source’s extension symbol $C(\sigma_e)$ with $C(\mathbb{T})$ in this specialization, and the standard implication from essential unitary equivalence of extensions to compact-unitary equivalence of the coordinate lifts.

References

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